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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE / B.TECH – IT /AIDS & M.TECH - CSE

20IT204 - COMPUTER ORGANIZATION AND ARCHITECTURE

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1 Which software translates high-level language to machine language?**
a) Linker b) Compiler c) Assembler d) Loader
2. **CO1 CPU execution time is given by _____.**
a) $CPI \times \text{Clock Rate}$ b) $(\text{Instruction Count} \times CPI) / \text{Clock Rate}$
c) $\text{Clock Rate} / CPI$ d) $\text{Instruction Count} / \text{Clock Rate}$
3. **CO2 The circuit that performs binary addition of two bits is called _____.**
a) Encoder b) Decoder c) Adder d) Comparator
4. **CO2 Bit-pair recoding improves Booth's algorithm by _____.**
a) Doubling the speed b) Reducing the number of steps by half
c) Eliminating additions d) Removing shifting
5. **CO3 Dynamic RAM (DRAM) stores data using _____.**
a) Flip-flops b) Capacitors c) Magnetic cores d) Transistors only
6. **CO3 The performance of cache is measured by _____.**
a) Hit ratio b) Miss ratio c) Access time d) All of the above

7. **CO4** In a processor, the Instruction Set Architecture (ISA) defines _____.
 a) Hardware design details only b) Software applications
 c) Interface between hardware and software d) Memory technology used

Fill in the blanks: (3x 1 = 3 Marks)

8. **CO1** The data on which operations are performed are called _____.
 9. **CO4** The first step in instruction execution is _____.
 10. **CO6** _____ law gives the theoretical speedup of a program using multiple processors.

PART - B (5x 3 = 15 Marks)

11. **CO1** Define Moore's Law.
 12. **CO2** Perform subtraction using 2's complement method: A = 1101 B = 1010
 Find A – B
 13. **CO6** Write the main advantage of multicore processors.
 14. **CO4** Explain the role of the control unit.
 15. **CO5** List the five stages of a typical instruction pipeline.

PART - C (5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Define Instruction Set Architecture. Discuss its components and importance. (8)
CO1 ii) Compare and contrast RISC and CISC architectures. (7)
 17. **CO2** i) Calculate Floating Point Representation and perform normalization and arithmetic operations for the following value -5.75 (7)
 IEEE 754 Single Precision Format (32 bits)
CO2 ii) Multiply M = +5 (0101) by Q = -3 (1101) using 4-bit Booth's algorithm. (8)
 Show all steps.
 18. **CO2** i) Calculate the non-restoring division for the following Dividend 13 (8)
 (1101₂) by Divisor 3 by 4-bit registers.

- CO2** ii) Explain addressing modes with suitable example and diagrams. (7)
19. **CO3** Explain the concept of Memory Mapping and its types with a neat diagram. (15)
20. **CO4** Describe the design of a data path and control unit for instruction execution with a neat diagram. (15)
21. **CO5** Explain pipeline hazards and their solutions with examples. (15)
22. **CO5** Explain Tomasulo's Algorithm for Dynamic Scheduling with diagram. (15)
